

Quick Comparison of Improv's Jazz DSP to Oak DSP

The Jazz DSP is a general-purpose, configurable VLIW architecture that can be configured into a custom processor optimized for a specific application. The Oak DSP (now called the Teak-Lite) is a fixed 16-bit general-purpose DSP. For the purposes of this discussion, we will compare the Oak to a fixed 16-bit Jazz DSP that is configured to have identical features to the Oak. It should be noted that the Jazz DSP can be additionally configured to a) increase parallelism and computation units, b) add custom instructions, c) change memory interfaces and registers and d) add or remove processor resources.

In general, it is possible to configure a Jazz DSP that has an identical feature set to the Oak DSP including capabilities (arithmetic, MAC, etc.), performance (up to 200 MHz 0.13um worst case), implementation (both synthesizable), interrupt support and on-chip memories. However, there are subtle differences that give the Jazz a real advantage over the Oak even with identical configurations.

- Computation/Address Units – The Oak DSP has two units: Computation and Bit Manipulation Unit (CBU) and Data Address and Arithmetic Unit (DAAU). The Jazz DSP has separate ALU, MAC, SHIFT and 2-3 Memory Interface Units (MIU).
- Register Availability – By having multiple units, the Jazz DSP has the advantage that each unit holds its own registers and therefore does not require movement of data between operations in different units significantly increasing performance (by reducing data movement instructions) within the processor.
- Ability to Parallelize – Even keeping the same number of computation units, it is simple to change the Jazz DSP to be a 2 slot (2 instructions per clock cycle) processor that would add minimal gate/instruction overhead but achieve up to 2X performance on the same software. This could be used to increase performance or reduce power (by lowering clock rate). It also means that a customer who is using the full performance of the DSP (190-200 MHz) can reduce the clock rate down to 100 MHz to ease implementation and programming.
- Programming – Both the Jazz and Oak can be programmed in C (Jazz additionally supports Java-based DSP programming). The Jazz also includes a broad collection of 'built-ins' that give the programmer access to low level operations and memory functions without having to move into assembly programming.
- Compiler Performance – The Jazz DSP has significant compiler optimizations built into the compiler. In particular, it supports 'profile-directed compilation' where the compiler can read profiled results of simulation and improve compilation for subsequent runs.

We have done conversion from Oak DSP C code for a previous customer involved in the Voice-Over-IP domain (simple running G.711 and G.726 with G.168 line-echo cancellation). With an identical processor, we were able to achieve a 15% reduction in clock rate requirement and by moving to a 2-slot configuration we achieved 40% reduction. Die size was essentially the same. Unfortunately, we can not share this code since it was proprietary to the customer.

To convert the code, we would cut and paste the C code into a task description, create some structure around the task interface, and convert the test bench to run in our environment. We would then compile to a target processor and profile the code using our Jazz Profiler tool.